Speech Technologies

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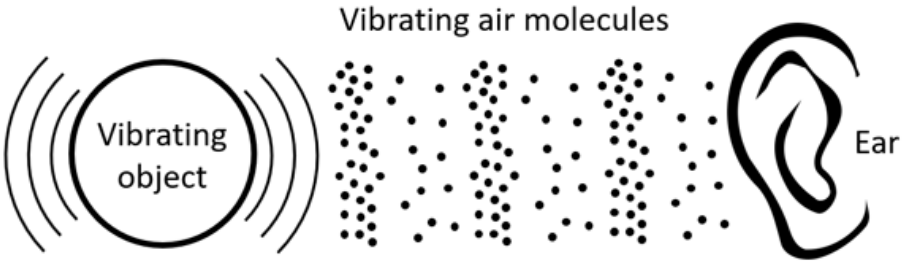
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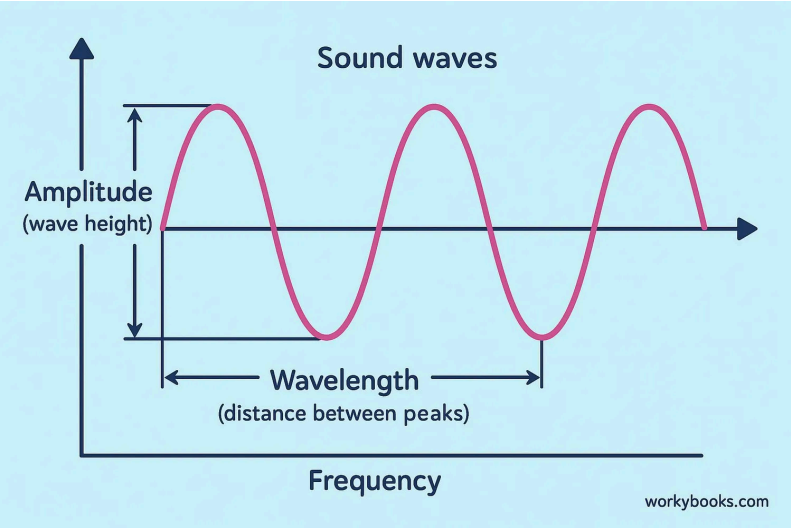
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1. Digital Speech Processing

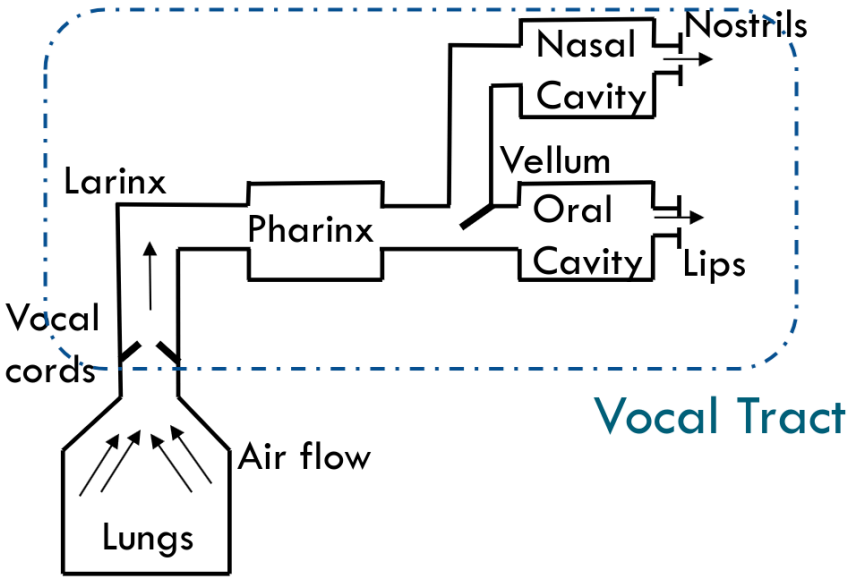
- **Sound**: A product of object vibration and these vibrations create changes in the air pressure propagating through space.



We can visualize soundwaves on a graph where the x axis is for the time and the y axis is for the pressure.

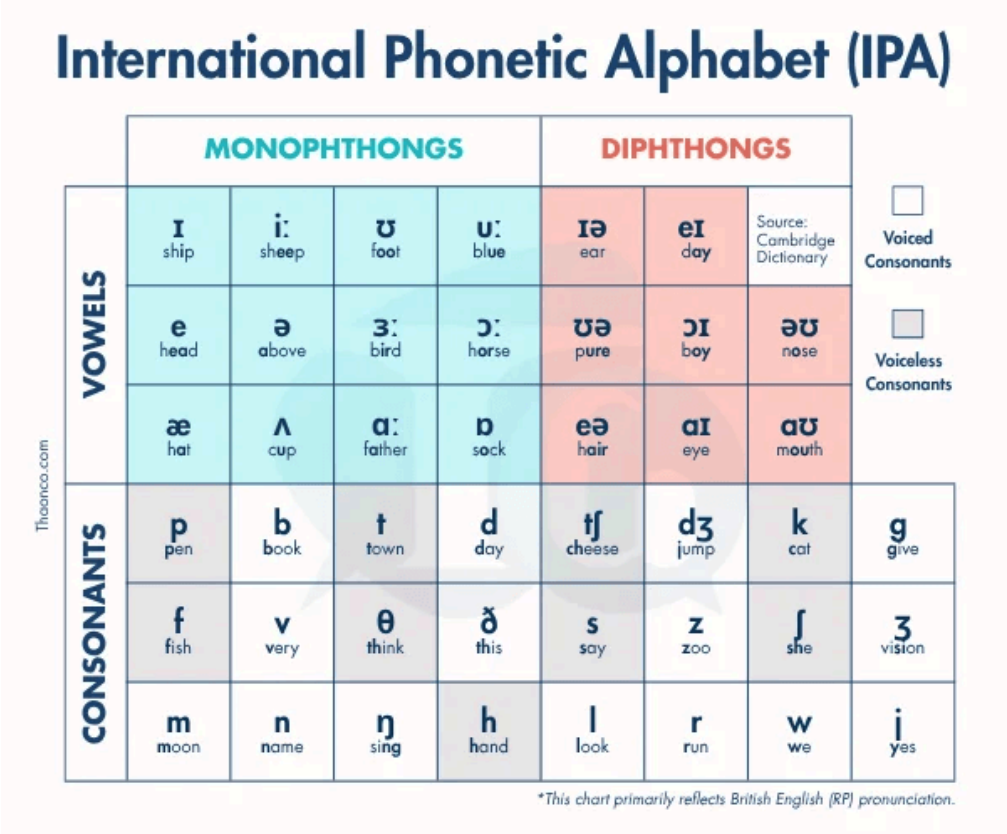


- **Speech Production**:
  1. Air flow is generated by the lungs.
  2. Vocal cords can vibrate or let the air through.
  3. Then the sound is modulated through the articulators (jaw, tongue, velum, lips and mouth) which change the shape of the vocal tract.
  4. The speech is propagated from lips and nostrils.



- **Co-articulation**: Speech sound influence eachother during speech production meaning that sounds are not separated.

- **Phonemes**: It's the smallest unit of sound in a language that helps us distinguish different words.
  - **Phones**: Inside a single phoneme category there are different speech sounds used in language, those are the phones, in fact if you swap phonemas from a word the meaning changes but if you do that with a phone it won't. Phones have consistent *spectro-temporal characteristics* if pronounced in isolation.
- Phonemas are usually classified in:
  - **Vowels**: They are produced with vibration of the glottis and articulated without major constraints.
  - **Consonants**: If pronounced alone they will result in constrictions in the throat or obstruction in the mouth, which means they can't be pronounced without a vowel.
- They can be categorized also by their **manner of articulation**:
  - **Occlusions**: p, b, t.
  - **Fricatives**: s, z.
  - **Nasals**: m, n.
  - **Liquids**: r.
  - **Affricates**: ts, tz.
- **International Phonetic Alphabet (IPA)**: It's an alphabetics system of phonetic notations based on the latin script, it's the standard for speech sound representation.
- We can classify speech sounds into two categories:
  - **Voiced**: Vibration of the vocal cords (both with vowels and consonants), it generates a *periodic signal* that holds a *fundamental frequency*.
  - **Unvoiced**: No vibration of the vocal cords, which results in no periodicity in the audio waveform.
- **Place and Manner of articulation**: They determine a specific configuration of the vocal tract that controls the air flow.



2. Automatic Speech Recognition

3. Speaker Diarization

4. Multimodality and LLMs

5. Training a Large-scale Model

6. Low Resourced Languages and Domains

7. Distant ASR